When Verbalization Is Not Introspection: Causal Tests for Internal-State Reporting

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Abstract

Recent work suggests that some language-model representations are especially accessible for verbal report, raising the possibility that models could be trained to faithfully describe their own internal states. We test whether such verbalization reflects genuine internal access or merely reconstruction from correlated input cues. Across GPT-2 (124M), Qwen2-0.5B-Instruct, and Llama-3.1-8B-Instruct, we causally edit concept directions at intermediate layers and train lightweight LoRA adapters to report the edited concept. Under standard correlated training, all three models achieve near-perfect reporting accuracy and generalize across unseen layers, positions, and intervention strengths. However, this apparent success collapses under a *decorrelation* test in which prompt content conflicts with the edited direction: direction-following accuracy falls to 0%, input-following reaches 100%, and the direction-following index is -1.0 for all three models. Ablation experiments further show that removing the edited direction does not affect reporting accuracy, indicating that the learned verbalization channel is non-causal. Llama-3.1-8B additionally collapses held-out concepts onto nearby trained concepts, suggesting generalization through representational similarity rather than semantic content. In contrast, a safety-oriented prompt-injection experiment with semantically varied training examples produces direction-following behavior on held-out phrasings, with particularly strong causal evidence in Llama-3.1-8B. These results show that high verbalization accuracy alone is insufficient evidence of introspection. We argue that *decorrelated interventions and causal ablations are necessary tests for faithful internal-state verbalization*.

1 Introduction

Language models contain rich internal representations that can often be identified using probes, sparse autoencoders (SAEs) [1], and other interpretability methods. However, just because some information can be decoded from a model does not mean the model can directly use or verbalize it. Gurnee et al. [2] study this difference using the Jacobian lens (J-lens) and identify a set of verbalizable representations, called the *J-space*. Information in this space can be reported, modified, and used in later reasoning. This raises a simple question: can we train a model to verbalize internal representations that are not naturally accessible? One approach is to set an internal direction to represent a concept c and train the model to report that concept. However, there is an important problem. During training, the internal edit, the prompt, and the target answer are usually correlated. As a result, the model may learn to guess the answer from the prompt instead of actually reading the edited internal state.

We test this problem using a causal *decorrelation test*. For an internal direction that represents two concepts, X and Y, we vary the internal direction and the prompt independently. This creates cases where both agree and cases where they conflict. The conflict cases are the key test: if the model truly

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reads the internal direction, it should follow the edited state rather than the prompt. We evaluate this setup on GPT-2, Qwen2-0.5B-Instruct, and Llama-3.1-8B-Instruct.

Our main result is simple but important. Under standard correlated training, all three models appear to learn the task very well and reach 100% accuracy on training-like examples. However, when the prompt and the edited direction disagree, all three models follow the prompt instead of the internal direction. Causal checks support the same conclusion: removing the edited direction does not reduce reporting accuracy, while removing input information does. This suggests that the models learn a shortcut from the prompt rather than a true verbalization channel. We also find an additional failure mode in Llama-3.1-8B-Instruct, where held-out concepts collapse onto the nearest trained concept pair, suggesting that the learned channel may rely on representational similarity rather than the intended semantic meaning.

Our contributions are:

1. We introduce a simple **direction–input decorrelation test** and a direction-following index (DFI) to distinguish true internal-state reporting from prompt-based shortcuts.
2. Across three models from 124M to 8B parameters, we show that **high reporting accuracy can give false evidence of introspection**.
3. We show that **causal tests are necessary**: direction ablation, input ablation, and Jacobian-alignment analysis help reveal whether the report actually depends on the edited internal state.
4. We identify a **concept-collapse failure** in Llama-3.1-8B-Instruct, while also showing that a safety-oriented prompt-injection experiment can produce genuinely direction-following behavior.

2 Method

2.1 Models and internal directions

We study three models: GPT-2 small [3], Qwen2-0.5B-Instruct [4], and Llama-3.1-8B-Instruct [5]. GPT-2 provides a small and well-studied setting for mechanistic interpretability [6], while Qwen2 gives us a different architecture and an instruction-tuned model that can respond more naturally to introspective prompts. Llama-3.1-8B-Instruct extends the study to a much larger model and requires layer-specific calibration of intervention strength, since the same coefficient does not work equally well across all layers (Section 2.2).

For each model, we build a bank of internal directions at selected residual-stream layers. The bank contains five types of directions: (i) J-lens directions, (ii) SAE decoder directions, (iii) linear-probe directions, (iv) norm-matched random directions, and (v) random directions restricted to the SAE decoder space. We also measure the causal effect of each direction by checking how steering along it changes the model's next-token prediction. In the experiments reported here, we mainly use named J-lens concept directions because they provide clear ground-truth labels for what the model should report. Our J-lens implementation is based on the public description of the method rather than the authors' original implementation, so we use it as a controlled way to construct concept directions and do not claim to exactly reproduce the original J-space.

For GPT-2, we use layers $2, 4, 6, 8, 10$ and the concepts Paris, Rome, dog, happy, sad, France, Italy. For Qwen2, we use layers $4, 8, 12, 16, 20$ and additionally include *cat*. For Llama-3.1-8B-Instruct, we use layers $3, 7, 19, 23, 27$ and the same eight concepts as Qwen2. The Llama direction bank contains 840 directions in total: 40 J-lens, 200 SAE, 200 probe, 200 random, and 200 random-in-SAE-span directions. We estimate the averaged Jacobian using 300 background contexts.

2.2 Activation editing

Let h_ℓ be the residual-stream activation at layer ℓ, and let d be a unit concept direction. We intervene by removing the current component along d and setting it to a controlled value α:

$$h'_\ell = h_\ell - \langle h_\ell, d \rangle d + \alpha d. \tag{1}$$

For a bipolar pair (X, Y) we define a signed direction $d_{X,Y}$; positive α corresponds to X and negative α to Y. The main training magnitude is $|\alpha| = 8$ for GPT-2 and $|\alpha| = 6$ for Qwen2, based on model-specific pilot sweeps.

For Llama-3.1-8B-Instruct, a single flat magnitude does not transfer across the tested layer range. A per-layer coherence sweep (varying α from 0.5 to 24 and scoring free-text coherence and exact target-concept mention) reveals two qualitatively distinct regimes: at layers 3 and 7, generations hit a coherence ceiling around $\alpha \approx 1.5$–2 and never verbalize the target concept cleanly at any tested magnitude, whereas layers 19, 23, and 27 remain coherent up to $\alpha \approx 6$–7 and reliably verbalize the target beyond that point. This contradicts the intuitive assumption that steering magnitude should scale smoothly with layer depth or hidden-state norm. We therefore calibrate α per layer: $\{3{:}1.5,\ 7{:}1.5,\ 19{:}6.0,\ 23{:}6.0,\ 27{:}6.5\}$, and use this table wherever Llama-3.1-8B results below reference "the calibrated α."

2.3 Native verbalizability

Before training, we clamp directions and measure whether free-text reports match the target concept. GPT-2's full direction bank contains 835 directions: 35 J-lens, 200 SAE, 200 probe, 200 fully random, and 200 random-in-SAE-span directions. Exact-match scoring is directly defined only for named J-lens directions; SAE, probe, and random directions require semantic annotation and are not used for the headline exact-match claim. We additionally compute three covariates for each direction: Jacobian magnitude, unembedding alignment, and workspace participation.

2.4 Inducing a verbalization channel

We train lightweight LoRA adapters only on *correlated* examples, where the edited direction, prompt content, and target report agree. For GPT-2, we train on (Paris, Rome), (dog, cat), and (sad, happy) at layers $\{2, 4, 6, 8\}$, holding out France/Italy as an unseen direction and layer 10 as an unseen layer. For Qwen2, we train on (Paris, Rome), (dog, cat), and (sad, happy) at layer 4, holding out France/Italy and layer 8. Llama-3.1-8B-Instruct uses the same three training pairs at layers $\{3, 7, 19, 23\}$, with a separately calibrated intervention strength α for each layer; France/Italy and layer 27 are held out.

Training examples are reshuffled at every epoch. This is especially important for Llama-3.1-8B-Instruct, where a fixed ordering causes strong interference between concept pairs and leads to near-zero evaluation accuracy despite low training loss. With shuffling, training is stable: the loss falls from $\mathcal{O}(1)$ to below 10^{-5} within 10 epochs, and evaluation reaches a 100% match rate.

We train only on correlated examples so that performance on later conflict conditions cannot be explained by prior exposure to cases where the prompt and edited direction disagree.

2.5 Decorrelation test

For each concept pair (X, Y) we independently manipulate direction state and input content:

	Input X	Input Y
Direction X	correlated	**conflict**
Direction Y	**conflict**	correlated

$$\begin{aligned}\mathrm{DFI} = \tfrac{1}{2}\,[&P(r = X \mid d = X, i = Y) - P(r = Y \mid d = X, i = Y) \\ &+ P(r = Y \mid d = Y, i = X) - P(r = X \mid d = Y, i = X)]\end{aligned} \tag{2}$$

where r is the report, d the edited direction, and i the input cue. $\mathrm{DFI} = 1$ indicates perfect direction following; $\mathrm{DFI} = -1$ indicates perfect input following.

2.6 Mechanism checks

Behavioral conflict tests are supplemented by two causal checks. First, we compare report accuracy after *direction ablation* and *input ablation*. A faithful channel should depend on the edited direction; a reconstruction channel should depend on the input. Second, we measure pre/post-training alignment between the edited direction and the downstream Jacobian associated with the target report. This is a diagnostic rather than a sufficient criterion: increased alignment would be consistent with a newly installed route from the direction to verbal output, while unchanged or diminished alignment weakens that interpretation.

2.7 Safety capstone: covert prompt-injection state

Finally, we construct a task-specific bipolar J-lens direction contrasting *injection* and *safe*. A fresh LoRA adapter (never warm-started from the main induction adapter) is trained on multiple correlated prompt-injection and benign scenarios, then tested on held-out phrasings in the two decorrelated conflict cells. This experiment asks whether a verbalization channel can succeed when the internal variable is safety-relevant and the training data contains semantic variation rather than a small set of unrelated concept labels. For Llama-3.1-8B-Instruct the capstone direction is built at layer 19 with the calibrated $\alpha = 6.0$, using the same correlated-only recipe as the main induction experiment.

3 Results

3.1 Native self-report is weak

GPT-2 shows almost no native ability to report the edited internal direction under our fixed elicitation prompt. Across all 35 named J-lens directions in the direction bank, exact-match accuracy is 0%. Its generations often refer broadly to "thinking" or "concepts" but do not identify the clamped target. We interpret this mainly as a capability limitation of a small, non-instruction-tuned 124M model, rather than evidence that the edited information is absent.

The pre-training decorrelation results support this interpretation. Direction-match accuracy is 0% for GPT-2 and 2.1% for Qwen2, while input-match accuracy is 5.0% and 22.9%, respectively (Table 2). Thus, before LoRA training, neither model has a reliable free-text channel for reporting the edited direction.

Despite this weak verbalization ability, GPT-2's direction-bank statistics clearly separate J-lens directions from the controls. Mean unembedding alignment is 0.569 for J-lens directions, compared with 0.201 for SAE directions, 0.114 for probe directions, 0.130 for random directions, and 0.138 for random directions restricted to the SAE span. Mean workspace participation is also close to 1.00 for J-lens directions, compared with only 0.08–0.11 for the other direction classes. These results suggest that the direction bank captures the intended J-lens positive controls, although our J-lens extraction remains approximate.

Llama-3.1-8B-Instruct shows a stronger but highly layer-dependent pattern. Mean causal efficacy decreases steadily with depth for J-lens directions, from 9.25 at layer 3 to 7.39, 3.53, 2.76, and 1.57 at layers 7, 19, 23, and 27, respectively. SAE, probe, and random-control directions show the same overall trend (Appendix A.1). A simple explanation is that perturbations inserted at earlier layers pass through more downstream computation and therefore have a larger effect on the final logits. We do not observe any unusual class- or layer-specific spike that would suggest the coherence-collapse regime discussed in Section 2.2.

Before training, Llama-3.1-8B-Instruct already shows some ability to follow the edited direction. Its direction-match accuracy is 35.8%, compared with an input-match accuracy of 25.0% ($n = 20$; four concept pairs across five layers), giving a mean DFI of $+0.11$ (Table 2). However, this average hides a clear layer-dependent transition. DFI is negative at layers 3 and 7, where the prompt input dominates, but becomes positive at layers 19, 23, and 27, where the model increasingly follows the edited direction.

Interestingly, the same transition between layers 7 and 19 also appears independently in the logit-lens analysis and in the coherence sweep of Section 2.2. Together, these three pre-training measurements point to a similar depth range where the edited representation becomes substantially more verbalizable.

3.2 Correlated training looks highly successful

After LoRA training, all three models achieve 100% accuracy on correlated, training-like evaluation examples. Standard generalization tests also give encouraging results for GPT-2 and Qwen2 (Table 1). GPT-2 achieves 100% mean match rate on unseen intervention strengths and on the held-out layer, while Qwen2 achieves 83.3% on unseen strengths and 100% on the held-out layer. For the unseen France/Italy direction, GPT-2 reaches 41.7% mean exact match, whereas Qwen2 reaches 100% in both tested settings.

Table 1: Post-training generalization under *correlated* evaluation. These results appear strong but do not establish that the report is caused by the edited direction.

Evaluation	GPT-2	Qwen2-0.5B-Instruct	Llama-3.1-8B-Instruct
Training-like correlated accuracy	100.0%	100.0%	100.0%
Unseen direction pair	41.7% ($n = 8$)	100.0% ($n = 2$)	0.0% ($n = 8$)
Unseen layer/position	100.0% ($n = 6$)	100.0% ($n = 6$)	100.0% ($n = 6$)
Unseen coefficients	100.0% ($n = 120$)	83.3% ($n = 30$)	82.8% ($n = 120$)

Table 2: Direction–input decorrelation. A positive DFI indicates direction following; a negative DFI indicates input following. Post-training reports follow the input perfectly in all three models.

Model	Setting	Acc. vs. direction	Acc. vs. input	DFI
GPT-2	Pre-training	0.0%	5.0%	-0.05
	Post-training	**0.0%**	**100.0%**	$\mathbf{-1.00}$
Qwen2	Pre-training	2.1%	22.9%	-0.21
	Post-training	**0.0%**	**100.0%**	$\mathbf{-1.00}$
Llama-3.1-8B	Pre-training	35.8%	25.0%	$+0.11$
	Post-training	**0.0%**	**100.0%**	$\mathbf{-1.00}$

Llama-3.1-8B-Instruct generalizes well to unseen layers and, at some layers, to unseen intervention strengths. The held-out layer 27, which is never used during training, achieves 100% mean match rate across all three trained concept pairs ($n = 6$). Robustness to unseen intervention strengths depends strongly on layer depth. Layers 19 and 23 remain perfectly accurate for all tested values from $\alpha = 4$ to $\alpha = 16$ (100%, $n = 60$). In contrast, layers 3 and 7 become less reliable as the intervention strength moves beyond the calibrated value, with mean match rates of 46.7% and 84.4%, respectively ($n = 30$ each). Overall accuracy across unseen intervention strengths is 82.8% ($n = 120$). These are also the shallow layers identified as having lower coherence in Section 2.2.

Generalization to an unseen *direction*, however, fails completely for Llama-3.1-8B-Instruct. On the held-out France/Italy pair, exact-match accuracy is 0% across all eight tested layer–sign combinations. The errors are highly systematic rather than random: every positive intervention is reported as "Paris," while every negative intervention is reported as "Rome," regardless of the layer.

This suggests that the adapter does not learn a general rule for reading arbitrary direction content. Instead, it appears to learn mappings tied to the direction vectors seen during training. Because France/Italy is semantically related to the trained Paris/Rome pair, its representation may lie close to Paris/Rome in activation space. The adapter therefore maps the unseen France/Italy direction to the nearest familiar pair rather than correctly reporting its actual content.

If we looked only at these generalization results, we might conclude that the adapters have learned to read the edited internal state. However, the decorrelation test leads to a different conclusion. The failure on the held-out direction in Llama-3.1-8B-Instruct already provides an important clue: the learned reporting channel appears to generalize to nearby familiar direction patterns, or *attractor basins*, rather than to the underlying semantic content of the edited direction itself.

3.3 Decorrelation exposes complete shortcut learning

Table 2 shows the key conflict-cell result. After training, direction-following accuracy is 0% for all three models, while input-following accuracy is 100%. Accordingly, the mean DFI is -1.0 in every case.

For example, when the Paris direction is clamped but the input suggests Rome, the model reports *Rome*. When the Rome direction is clamped but the input suggests Paris, it reports *Paris*. The same behavior appears across all tested concept pairs and across models ranging from 124M to 8B parameters.

Table 3: Post-training causal ablations. If reports faithfully read the internal edit, direction ablation should hurt more than input ablation. We observe the opposite in all three models.

Metric	GPT-2	Qwen2	Llama-3.1-8B
Baseline report accuracy	100.0%	100.0%	100.0%
After direction ablation	100.0%	100.0%	100.0%
After input ablation	50.0%	0.0%	50.0%
Direction necessary	0/8	0/2	0/8
Input necessary	4/8	2/2	4/8

The gap between correlated and decorrelated evaluation is therefore 100 percentage points for all three models. This is a clear sign of shortcut learning: the models learn to perform the reporting task during training, but they do not learn to rely on the internal direction that we intended them to report.

Llama-3.1-8B-Instruct is particularly interesting because it is the only model with a positive *pre-training* DFI ($+0.11$). This effect comes mainly from layers 19, 23, and 27, where the base model already shows some sensitivity to the edited direction (Section 2.1). However, correlated-only LoRA training does not strengthen this behavior. Instead, it removes it: after training, DFI falls to -1.0 across all four trained layers, including layers 19 and 23, where the untrained model previously followed the direction.

Section 3.4 provides additional evidence for this reversal, showing a corresponding drop in Jacobian alignment at the same layers.

3.4 Ablations confirm that reports are input-driven

The mechanism checks provide further evidence for post-training causal ablation (Table 3). For GPT-2, baseline report accuracy is 100%. Removing the edited direction does not change accuracy, which remains at 100%, while removing the input reduces accuracy to 50%. Across the eight tested settings, the edited direction is necessary in 0% of cases, whereas the input is necessary in 50%.

Qwen2 shows an even clearer result. Direction ablation again leaves accuracy unchanged at 100%, while input ablation reduces accuracy to 0%. Thus, in both tested settings, the input is necessary for the report but the edited direction is not.

Llama-3.1-8B-Instruct shows almost the same pattern as GPT-2. Baseline accuracy is 100%, direction ablation leaves it unchanged at 100%, and input ablation reduces it to 50%. The direction is necessary in none of the eight tested pair–layer combinations, while the input is necessary in four of eight.

Interestingly, this effect differs across concept pairs. For Paris/Rome, removing either the direction or the input does not change the report: the model continues to output "Paris" or "Rome." For dog/cat, however, removing the input changes the report. This is consistent with the concept-collapse result above. In particular, "Paris" and "Rome" appear to act as default outputs for positive and negative steering. This may also explain why the unseen France/Italy direction is incorrectly mapped to Paris/Rome.

Jacobian alignment also fails to show the expected installation of a verbal route. For GPT-2, mean alignment changes from 0.0290 before training to approximately zero after training ($\Delta = -0.0290$), increasing in only 1/8 tested settings. Qwen2 changes from -0.0160 to approximately zero ($\Delta = +0.0160$), increasing in 1/2 settings. Llama-3.1-8B-Instruct changes from -0.0273 to -0.0640 ($\Delta = -0.0367$), increasing in 4/8 tested settings; broken down by depth, alignment *increases* post-training at the shallow layers (3, 7: $+0.196$ and $+0.328$ for Paris/Rome, $+0.323$ and $+0.074$ for dog/cat) but *decreases* at the deep layers (19, 23: -0.062 and -0.281 for Paris/Rome, -0.361 and -0.510 for dog/cat) precisely the layers where Table 2 shows the pre-training DFI was positive and training drove it to -1.0. The signs are not directly comparable across models, but no result across any of the three models supports a simple story in which LoRA training strengthens a stable causal map from the edited direction to the verbal report; if anything, the Llama-3.1-8B breakdown shows training *removing* sensitivity specifically where the base model already had some.

Table 4: Held-out prompt-injection capstone ablations for Llama-3.1-8B-Instruct. Values are report match rates under different interventions; `no_hook` is a control with no steering hook applied at all.

Split	Target	Baseline	Ablate direction	Ablate input	No-hook control
train	injection	1.00	0.33	1.00	0.67
train	safe	1.00	0.33	1.00	1.00
held-out	injection	1.00	0.00	1.00	0.00
held-out	safe	1.00	0.67	1.00	1.00

3.5 A safety-specific positive case, with mechanistic support that strengthens at scale

The prompt-injection experiment gives a very different result from the generic concept experiment. We train a new adapter to report either *injection* or *safe* from a bipolar internal direction, using several semantically different scenarios. On held-out scenario phrasings, all three models achieve perfect direction following in the conflict cells, with $\mathrm{DFI} = +1.0$. Thus, unlike the generic concept task, the safety-specific monitor follows the edited direction rather than simply repeating the conflicting prompt content.

However, the mechanism checks show that the two smaller models should be interpreted cautiously. For GPT-2, the held-out *injection* condition depends partly on the edited direction: direction ablation reduces accuracy from 1.0 to 0.67. In contrast, the held-out *safe* condition is unaffected, and some conditions can be solved simply by the presence of the hook. Qwen2 shows the opposite asymmetry. Its *safe* condition depends strongly on the direction, with accuracy dropping from 1.0 to 0.0 after direction ablation, while the *injection* condition remains correct even when the direction is removed or the hook is absent. For both models, Jacobian alignment does not consistently increase and instead moves closer to zero.

Llama-3.1-8B-Instruct provides the strongest causal evidence (Table 4). Direction ablation reduces accuracy in all four tested conditions: train/held-out $\times$ injection/safe gives $1.00 \to 0.33$, $1.00 \to 0.33$, $1.00 \to 0.00$, and $1.00 \to 0.67$, respectively. Thus, the edited direction is necessary in every tested condition.

The held-out *injection* case is especially clean. Removing the edited direction and removing the steering hook entirely produce exactly the same result: both give 0% accuracy and both cause the model to report "safe." In contrast, ablating the input leaves accuracy at 100%. This suggests that the edited direction, rather than the prompt content or a default output bias, is responsible for the correct *injection* report.

The *safe* condition is less clean because the base model already tends to output "safe": the no-hook control achieves 100% accuracy on both train and held-out examples. Even so, direction ablation still reduces accuracy from 1.00 to 0.33 on training examples and from 1.00 to 0.67 on held-out examples, showing that the direction still makes a causal contribution.

Jacobian alignment changes from $+0.0281$ to -0.0501 ($\Delta = -0.0782$), moving toward zero rather than supporting the ablation result. This again shows that Jacobian alignment alone is not a reliable measure of causal grounding. In this setting, the decorrelation and ablation tests provide the stronger evidence.

Overall, we view GPT-2 and Qwen2 as behavioral positive cases with only partial and asymmetric causal support. Llama-3.1-8B-Instruct provides a much stronger positive result, where successful direction following is also supported by the mechanism checks. Still, we treat these checks as necessary evidence rather than considering any single successful run as conclusive.

4 Limitations

Our experiments should be viewed as a proof of methodology rather than a definitive statement about introspection in frontier language models.

First, GPT-2 small is not instruction-tuned and performs poorly on free-text introspection prompts, which creates a clear capability floor. Qwen2-0.5B-Instruct and Llama-3.1-8B-Instruct are better suited to the task, but they are still much smaller than current frontier models. Since we test only three

models, ranging from 124M to 8B parameters, we cannot determine whether the shortcut-learning behavior changes systematically with scale.

Second, our J-lens extraction is an approximate reimplementation using 300 averaging contexts. A closer reproduction of the original J-space method could produce somewhat different direction geometry.

Third, our main induction experiments use named J-lens concept directions because they provide clear ground-truth labels. The full direction banks also include SAE, probe, and random directions, including 840 total directions for Llama-3.1-8B-Instruct (Appendix A.1). However, evaluating free-text reports for these directions would require reliable human semantic annotation, which is not yet included in our cross-model experiments.

Fourth, the number of tested concept pairs is limited, especially for Qwen2. The capstone prompt-injection experiment is also restricted to the binary *injection*/*safe* distinction.

Fifth, our ablation interventions may introduce distribution shift or artifacts from the intervention hook itself. The capstone no-hook controls show that this can matter in some settings, including the Llama-3.1-8B-Instruct *safe* condition.

Finally, the causal-efficacy scores in the Llama-3.1-8B-Instruct direction bank are computed using a fixed sweep of $\alpha \in \{8, 16, 24\}$ rather than the layer-specific calibrated strengths used in the main experiments. We checked the layer-wise averages for signs of contamination from the coherence-collapse regime and found a smooth decrease with depth rather than an anomalous spike. However, we did not recompute the full metric using calibrated strengths. Importantly, this metric is not used in our main conclusions.

These limitations motivate future experiments with larger models, more diverse direction sets, broader concept coverage, and intervention methods that better preserve the model's original activation distribution.

5 Conclusion

We asked whether post-training can teach language models to faithfully report a controlled internal state. At first, the answer appears to be yes. GPT-2, Qwen2-0.5B-Instruct, and Llama-3.1-8B-Instruct all reach 100% accuracy on correlated training-like examples and generalize well to several standard held-out conditions. However, the causal tests give a different answer. When the prompt and edited direction disagree, all three models follow the prompt with 100% accuracy and follow the internal direction with 0% accuracy. Moreover, removing the edited direction does not reduce reporting accuracy. This shows that the learned channel is usually reconstructing the expected concept from external information rather than directly reading the manipulated internal state. Because the same result appears from GPT-2 at 124M parameters to Llama-3.1-8B-Instruct at 8B parameters, it is unlikely to be only a small-model limitation. Llama-3.1-8B-Instruct also reveals a second failure mode. When tested on an unseen concept direction, the model does not fail gradually; instead, it maps the new direction onto the nearest trained concept pair. These learned default outputs also help explain why ablating the direction sometimes has little effect. The prompt-injection experiment provides an important positive result. With richer, task-specific training, all three models can behaviorally follow the edited direction. For Llama-3.1-8B-Instruct, this behavior is also supported by strong ablation evidence showing that the internal direction is causally necessary. GPT-2 and Qwen2 show weaker and more asymmetric causal support. Overall, behavioral accuracy alone is not enough to establish faithful verbalization. We therefore propose a simple standard for future work: a verbalization channel should be considered faithful only if it continues to follow the internal state when external evidence is decorrelated, and if the resulting report is causally dependent on the internal state it claims to describe.

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A Experimental Details

A.1 Direction-bank composition

For GPT-2, the completed direction bank contains 835 directions: 35 J-lens directions across five layers and seven named concepts; 200 SAE decoder features; 200 probe directions; 200 fully random directions; and 200 random directions sampled within the SAE decoder span. The SAE release used by the pipeline is `gpt2-small-res-jb`, with residual-pre hook points. The Qwen2 configuration uses `Qwen/Qwen2-0.5B-Instruct`; its configured SAE is trained on base Qwen2-0.5B and is therefore an approximation when applied to the instruction-tuned model. Since the headline cross-model result is based on named J-lens directions, it does not depend on SAE semantic labels.

The Llama-3.1-8B-Instruct bank contains 840 directions across five layers $\{3, 7, 19, 23, 27\}$: 40 J-lens directions (eight named concepts $\times$ five layers), 200 SAE decoder features, 200 probe directions (drawn from sixteen relational natural-language prompt templates, e.g. "The native language of", "The twin city of", "The headquarters of", each scored with both a mean-difference and a logistic-regression probe direction), 200 fully random directions, and 200 random directions sampled within the SAE decoder span — 168 directions per layer, matching the class composition used for GPT-2. Table 5 reports mean causal-efficacy by direction class and layer; we use this only as a bank-validation statistic (Section 2.1) and not as a headline result.

Table 5: Mean causal-efficacy (KL divergence between steered and unsteered next-token distributions, averaged over contexts and $\alpha \in \{8, 16, 24\}$) by direction class and layer, Llama-3.1-8B-Instruct bank. Values decrease smoothly with depth for every class, with no anomalous layer-3/7 spike.

Direction class	Layer 3	Layer 7	Layer 19	Layer 23	Layer 27
J-lens	9.245	7.394	3.526	2.757	1.570
SAE	8.733	6.900	1.773	0.917	0.394
Probe	10.229	8.174	3.649	1.821	0.929
Random	7.074	5.175	1.337	0.700	0.303
Random-in-SAE-span	8.046	6.143	1.459	0.739	0.317

A.2 GPT-2 geometry summary

A.3 Training splits

GPT-2 trains on concept pairs (Paris,Rome), (dog,happy), and (sad,happy), at layers 2, 4, 6, and 8, holding out France/Italy and layer 10. Qwen2 trains on (Paris,Rome), (dog,cat), and (sad,happy) at layer 4, holding out France/Italy and layer 8. GPT-2 uses $|\alpha| = 8$ and Qwen2 uses $|\alpha| = 6$ for the principal induction experiments. Llama-3.1-8B-Instruct trains on (Paris,Rome), (dog,cat),

Table 6: Mean direction covariates in the GPT-2 direction bank.

Direction class	Jacobian magnitude	Unembedding alignment	Workspace participation
J-lens	0.0185	0.5686	1.0000
SAE	0.0097	0.2007	0.1021
Probe	0.0109	0.1142	0.0834
Random	0.0089	0.1299	0.0913
Random-in-SAE-span	0.0099	0.1382	0.1067

and (sad,happy) at layers $\{3, 7, 19, 23\}$, holding out France/Italy and layer 27, using the per-layer calibrated α given in Section 2.2 ($\{3 : 1.5,\ 7 : 1.5,\ 19 : 6.0,\ 23 : 6.0\}$ for the four trained layers; $27 : 6.5$ for the held-out layer).

A.4 Llama-3.1-8B-Instruct: unseen-coefficient robustness by layer

Table 7 breaks down the aggregate 82.8% ($n = 120$) unseen-coefficient result from Table 1 by layer, testing $\alpha \in \{4, 6, 10, 12, 16\}$ at both signs against all three trained concept pairs ($n = 30$ per layer). The break is sharp and matches the two-regime split identified in the calibration sweep (Section 2.2): layers 19 and 23 tolerate the full tested range without a single failure, while layers 3 and 7 degrade as α increases past their calibrated value, through the same two failure modes documented throughout the paper (repetition collapse and cross-concept drift).

Table 7: Llama-3.1-8B-Instruct mean match rate on unseen intervention coefficients, by trained layer ($n = 30$ per layer: 3 concept pairs $\times$ 5 magnitudes $\times$ 2 signs).

	Layer 3	Layer 7	Layer 19	Layer 23
Mean match rate	46.7%	84.4%	100.0%	100.0%

B Detailed Safety-Capstone Ablations

Table 8: Held-out prompt-injection capstone. Values are report match rates under different interventions.

Model	Target	Baseline	Ablate direction	Ablate input	No-hook control
GPT-2	injection	1.00	0.67	1.00	0.67
GPT-2	safe	1.00	1.00	1.00	1.00
Qwen2	injection	1.00	1.00	1.00	1.00
Qwen2	safe	1.00	0.00	1.00	0.00

The behavioral conflict test is perfect ($\mathrm{DFI} = 1$) in both models, but the ablations reveal asymmetric mechanisms. In particular, some labels remain correct when the semantic direction is removed or even when the hook is absent. We therefore do not treat these two models' Stage 7 results as conclusive evidence that a single faithful introspection channel has been installed. Llama-3.1-8B-Instruct's corresponding ablation table, covering both the training and held-out phrasing splits, is reported as Table 4 in the main text; unlike the two smaller models, direction ablation reduces accuracy below baseline for every tested target and split, and the held-out *injection* condition shows no separable hook-only or default-output confound.

Response to selection-ratio confound. We thank the reviewer for the observation that fixing the raw feature count at top-2048 across dictionary sizes confounds *selection ratio* with *dict_size*: 2048 features are 50% of a 4096-entry dictionary but only 3% of a 65536-entry one. We re-ran the overlap analysis holding the selection *ratio* fixed instead of the count: for each forget author and each dictionary size, we rank alive (non-dead) SAE features by forget–retain activation contrast and keep the top $\mathrm{round}(p \times n_{\text{alive}})$ features for $p \in \{10\%, \ldots, 100\%\}$, then compute Jaccard overlap between forget- and retain-active feature sets *within* that ratio-matched candidate set, at activation thresholds $\tau \in \{0, 0.01, 0.05, 0.10\}$.

Table 9: Percentage-matched Jaccard overlap (100% of alive features, $\tau = 0$) between forget- and retain-author active feature sets, across dictionary size, for all four models.

Model	**4096**	**8192**	**16384**	**32768**	**65536**
GPT-2 Small	0.960	0.964	0.965	0.968	0.964
GPT-2 Medium	0.902	0.8905	0.844	0.809	0.805
GPT-2 Large	0.902	0.891	0.871	0.826	0.801
Llama-3.2-1B	0.944	0.837	0.343	0.360	0.331

Table 9 reports mean Jaccard at $p = 100\%$, $\tau = 0$ across all five dictionary sizes for all four models in the paper, together with the Spearman correlation between dictionary size and Jaccard. Only Llama-3.2-1B shows a statistically significant decline ($\rho = -0.90$, $p = 0.037$); the three GPT-2 models remain essentially flat at $\tau = 0$, confirming that the original trend is not purely an artifact of the fixed-count confound. Notably, at non-zero thresholds ($\tau = 0.05, 0.10$) GPT-2 Large *also* develops a strong, significant decline ($\rho = -1.00$, $p < 0.001$), indicating that the dictionary-size effect is real but architecture- and threshold-dependent rather than uniform across models.